

Residual Heat Cascade: A Staged Architecture for Industrial Waste-Heat Reclamation

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Abstract

A large fraction of industrial energy input leaves sites as unused heat. Recovery equipment already exists—economizers, heat-recovery steam generators, organic Rankine cycles, supercritical CO₂ cycles, industrial heat pumps, absorption chillers, and thermal stores—yet much technically recoverable heat is still rejected. The usual failure is architectural rather than thermodynamic: a single recovery device is asked to serve every temperature band, or it competes with the host process for the same stream, or it is bypassed whenever load changes. This paper proposes the Residual Heat Cascade (RHC), a staged control and equipment architecture that allocates waste heat by quality. Process-critical heat is taken first. Only residual high-grade gas is admitted to a power cycle, preferably a partial-heating supercritical CO₂ or compact organic Rankine layout rather than a highly recuperated Brayton copy. Mid- and low-grade streams are upgraded with heat pumps or absorption chillers, or stored in sensible media, instead of being forced through a poor power cycle. A short-duration thermal buffer keeps the cascade online at part load. The architecture is developed first for gas-turbine-driven liquefied natural gas (LNG) trains, then generalized to compressor stations, refining, furnace industries, paper and chemicals, marine engines, and low-grade sites such as food plants and data centres. RHC does not claim a new working fluid or a violation of Carnot limits. It claims that matching quality, protecting host-process duty, and storing mismatch will convert more of the heat industry already buys into product, displaced fuel, or work.

Keywords: waste heat recovery; residual heat cascade; LNG liquefaction; supercritical CO₂; industrial heat pumps; thermal energy storage; process integration

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1 Introduction

Industrial processes convert only part of their purchased energy into useful service. The remainder leaves as stack gas, cooling water, radiation, and other reject streams. Estimates of unused industrial heat vary by method and region, but the scale is consistently large: on the order of thousands of thermal terawatt-hours per year globally, with a substantial share below 200 °C [McKinsey & Company, 2023, Forman et al., 2016, Bianchi et al., 2019]. McKinsey has argued that at least 3100 TWh_{th} of *feasible* waste heat remains uncaptured, with potential annual savings on the order of 140 billion euros if heat is reused as heat or converted to power where appropriate [McKinsey & Company, 2023].

The technology catalogue is not empty. High-temperature exhaust is routinely turned into steam. Organic Rankine cycle (ORC) units generate power from mid-grade sources. Industrial heat pumps now deliver hot water and, increasingly, low-pressure steam from lukewarm loops. Packed beds, concrete modules, molten salt, and hot-water pits can hold heat from hours to seasons. Surveys nevertheless list the same barriers year after year: capital cost per recovered unit of energy, variable or dirty sources, spatial mismatch between source and sink, lack of in-house know-how, and reluctance to attach unproven kit to a production-critical machine [Christodoulides et al., 2022].

A less advertised barrier is *mis-allocation*. A highly recuperated supercritical CO₂ (sCO₂) cycle can look efficient on a $T-s$ diagram and still leave flue gas far too hot. A steam plant can maximize electrical efficiency and starve an amine reboiler, forcing a fired heater back into service. An ORC can be installed on 90 °C aftercooler water and produce a few percent of Carnot work while a neighbouring boiler still burns gas. Operators then bypass the recovery train at part load because nothing stores the mismatch, and the “installed” system spends much of its life isolated.

This paper treats that allocation failure as the design problem. The Residual Heat Cascade is a rule set and a modular equipment train:

1. Never take heat the host process already needs.
2. Convert only residual high-grade heat to work, with a cycle chosen to cool the source rather than to maximize isolated cycle efficiency.
3. Upgrade or reuse mid- and low-grade heat as heat or cooling.
4. Store what cannot be used at the moment of rejection.
5. Account for product and displaced fuel, not generator kilowatt-hours alone.

Section 2 reviews the resource and the existing toolkit. Section 3 specifies the cascade. Section 4 gives a compact thermodynamic rationale. Section 5 applies the architecture to LNG liquefaction. Section 6 maps it onto other industries. Section 7 compares RHC with single-device strategies. Section 8 discusses packaging, controls, and risk. Section 9 states limits. Section 11 concludes.

2 Background

2.1 The resource is stratified

Waste heat is not one commodity. A gas-turbine exhaust at 425 °C to 550 °C can support work [Anonymous, 2024]. Jacket water at 90 °C can feed a heat pump or a hot-water network. Aftercooler loops near 40 °C are useful only if a low-temperature customer or a heat-pump evaporator exists nearby. European industry studies show that a large share of rejected heat sits at low temperature, while the *Carnot* (work) potential is concentrated in the hotter minority [Bianchi et al., 2019]. Any architecture that treats all reject streams as feedstock for electricity will systematically waste the larger, duller resource.

2.2 Existing devices and their blind spots

Process heat-recovery units. Hot-oil coils and heat-recovery steam generators (HRSGs) are the default first use of turbine and furnace exhaust. They are the correct first use whenever treating, drying, or steam headers depend on them. They are incomplete if the designer stops at the process duty and vents the remainder.

Steam Rankine bottoming cycles. On large industrial gas turbines, multi-pressure steam remains the highest-efficiency power option when water, space, and availability penalty are acceptable. Combined-cycle additions on liquefaction sites have been shown to lift power-island efficiency above 50 %, at the cost of footprint, water treatment, and a commonly cited availability penalty of one to three percentage points [Meher-Homji et al., 2015].

Organic Rankine cycles. ORC is the commercial power option for many mid-grade streams. It is water-free and automated. On gas-turbine waste heat it typically recovers less work than steam but more than a thermoelectric stack. Hydrocarbon working fluids can be an unwelcome fire load on methane sites.

Supercritical CO₂ cycles. Closed *s*CO₂ Brayton cycles are compact, dry, and well matched to exhaust above about 400 °C [Manente and Fortuna, 2020]. For waste-heat duty the layout matters. Strong recuperation raises cycle efficiency and reduces the temperature window available to cool the flue gas. Partial-heating and split-flow arrangements extract more heat from a finite stream even if isolated cycle efficiency is lower [Manente and Fortuna, 2020]. Commercial waste-heat *s*CO₂ is entering demonstration on pipeline compressor exhaust; it is not yet a standard liquefaction accessory.

Heat pumps and absorption chillers. Industrial heat pumps can lift lukewarm waste to process hot water or low-pressure steam. Absorption chillers convert mid-grade heat into cooling, including gas-turbine inlet-air cooling. These devices often beat power cycles on low-grade streams because they deliver heat or cooling services with coefficients of performance well above unity, whereas a heat engine on a small temperature difference cannot.

Thermal storage. Hot-water tanks, steam accumulators, packed beds, concrete modules, molten salt, and seasonal pits or aquifers can hold heat from minutes to months. Heat-to-heat round-trip efficiencies of 80 % to 95 % are routine for sensible stores. Cost per kilowatt-hour of *heat* is typically far below electrochemical batteries when the required output is thermal.

None of these devices is the cascade. Each becomes useful when placed on the correct band and prevented from cannibalizing a higher-priority duty.

2.3 Why single-device projects stall

Field surveys rank high capital cost per recovered energy unit and low or variable source quality as the leading technical reasons for not recovering heat [Christodoulides et al., 2022]. Organizational reasons include split incentives when heat is sold off-site, lack of metering, and production teams that will not accept backpressure, extra pressure drop, or a first-of-a-kind trip risk. RHC is written to those constraints: process-first interlocks, a published backpressure cap, modular blocks, and a buffer so the recovery train is not the first thing isolated.

3 Architecture of the Residual Heat Cascade

3.1 Definition

The Residual Heat Cascade is a four-stream allocation of a reject inventory, supervised by one controller, with a thermal buffer on the streams that would otherwise be dumped at part load.

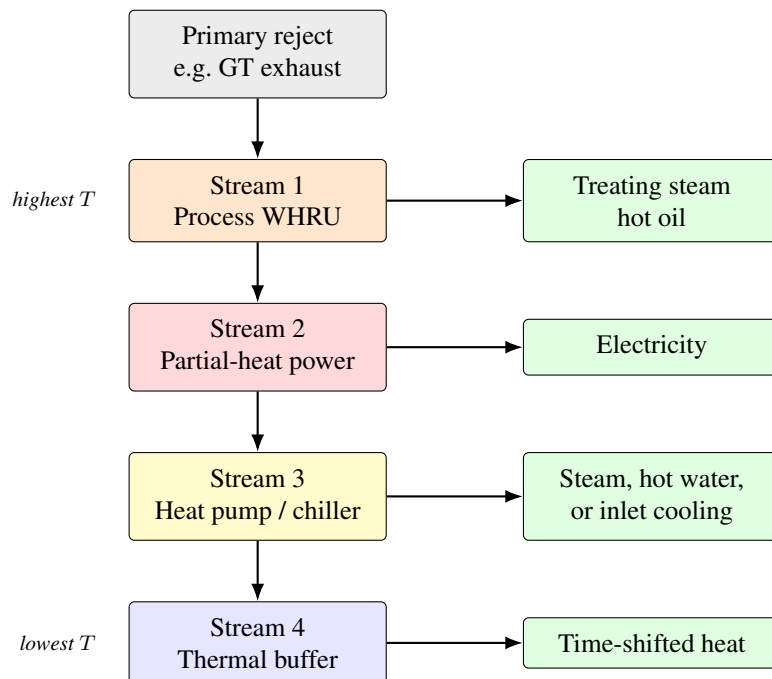


Figure 1: Quality-ordered Residual Heat Cascade. Each stage consumes only heat that remains after higher-priority duty.

3.2 Stream 1 — process-critical heat

The first heat exchanger is the host-process coil: amine reboiler, dehydration, kiln air preheat, paper-machine steam, or whatever duty would otherwise be fired. Control rule: Stream 2 may not reduce Stream 1 below its setpoint. If the process coil calls for more duty, the power block unloads first.

This rule is the difference between “recovery” and “theft.” A megawatt taken from an amine reboiler and put through a turbine is not a megawatt gained if a fired heater replaces it.

3.3 Stream 2 — residual work

Only after Stream 1 is satisfied is residual gas admitted to a power cycle. The recommended layouts are:

- a *partial-heating* or split-flow $s\text{CO}_2$ cycle when exhaust after the process coil remains above about 400 °C and compactness or water scarcity matters;
- a compact ORC when temperature, risk appetite, or vendor maturity favours a commercial package;
- steam Rankine when the gas turbine is large, water is available, and the owner already accepts HRSG complexity.

The objective function is net work from the *finite flue-gas stream*, subject to a stack-temperature floor (acid dew point, opacity, plume) and a gas-turbine backpressure cap. Isolated cycle thermal efficiency is a secondary metric.

3.4 Stream 3 — upgraded heat or cooling

Streams that leave Stream 2 still warm, together with compressor aftercoolers, jacket water, and similar loops, feed:

- a high-temperature heat pump delivering process hot water or low-pressure steam; or
- an absorption chiller, including turbine inlet-air cooling where ambient temperature limits driver power.

Inlet-air cooling on a mechanical-drive gas turbine increases available shaft power and can raise liquefaction or compression throughput. That product increment is often worth more than the electricity the same heat would produce in a low-temperature ORC.

3.5 Stream 4 — storage of mismatch

A two-tier store is specified:

- mid-grade sensible store (concrete modules, packed bed, or hot oil) in the approximate band 150 °C to 400 °C;
- low-grade hot-water tank below 100 °C.

Duration is hours, not seasons, unless the site exports to a district network. The purpose is operational: keep Streams 2 and 3 from being bypassed every time the host process swings.

Seasonal pits, aquifers, and borehole fields are optional addenda where a city or campus is the customer. They are not part of the default industrial skid.

3.6 Optional process-side squeeze

On cryogenic plants, residual Joule–Thomson valves in the cold box can be replaced by two-phase expanders. That measure is not heat recovery. It is listed because it stacks with RHC and converts leftover pressure drop into liquid product [Elliott Group, 2024].

3.7 Supervisory control

A single optimizer sits above vendor controllers. Inputs include process-heat demand, driver load, ambient wet-bulb or dry-bulb temperature, stack dew-point margin, measured backpressure, store state of charge, and electricity or steam prices if relevant. Outputs are valve and bypass positions and store charge/discharge setpoints. Fail-safe is bypass to stack with process coil still in service.

4 Thermodynamic rationale

4.1 Finite-stream work is not cycle efficiency

For a flue-gas stream with mass flow $\dot{m}_g$, specific heat $c_{p,g}$, and inlet temperature $T_{g,\text{in}}$, the heat that can be extracted down to a permitted stack temperature $T_{g,\text{out}}$ is

$$\dot{Q}_{\text{avail}} = \dot{m}_g c_{p,g} (T_{g,\text{in}} - T_{g,\text{out}}). \quad (1)$$

A power cycle converts a fraction η_{cyc} of the heat it actually absorbs, $\dot{Q}_{\text{abs}} \leq \dot{Q}_{\text{avail}}$:

$$\dot{W}_{\text{net}} = \eta_{\text{cyc}} \dot{Q}_{\text{abs}}. \quad (2)$$

Highly recuperated cycles raise η_{cyc} by preheating the working fluid, which raises the temperature at which the cycle can still accept heat and therefore raises $T_{g,\text{out}}$. The product $\eta_{\text{cyc}} \dot{Q}_{\text{abs}}$ can fall even as η_{cyc} rises. Waste-heat design therefore maximizes $\dot{W}_{\text{net}}$ subject to (1), not η_{cyc} alone [Manente and Fortuna, 2020].

4.2 Why low-grade heat should not default to power

The Carnot ceiling between a source at T_H and a sink at T_C is $1 - T_C/T_H$. For $T_H = 360$ K (87 °C) and $T_C = 300$ K, that ceiling is 17 %; real engines deliver much less. A heat pump using the same stream as evaporator source can deliver heat at a useful process temperature with a coefficient of performance of two to four under industrial conditions, displacing boiler fuel joule for joule more effectively than a

small ORC displaces grid electricity—especially where on-site power is already cheap relative to fired heat.

4.3 Accounting identity

The figure of merit proposed for RHC is avoided primary energy per unit of host product:

$$\Delta e = \frac{E_{\text{fuel,base}} - E_{\text{fuel,RHC}}}{P_{\text{RHC}}} - \frac{E_{\text{fuel,base}}}{P_{\text{base}}}, \quad (3)$$

where P is product rate (tonnes of LNG, barrels, tonnes of clinker, and so on). Electricity exported is converted at the site's displaced heat rate. Inlet-chilling tonnes and expander tonnes are included in P_{RHC} . This identity prevents a project from looking successful on a generator meter while specific fuel use is unchanged.

5 Application to LNG liquefaction

5.1 Why liquefaction is a demanding test case

A typical large liquefaction plant consumes on the order of 280 kWh per tonne of LNG and may burn about 5 % of inlet gas as fuel, with a carbon intensity near 20 kg CO₂ per barrel of oil equivalent for conventional designs, lower for efficient electric-drive plants. Mechanical-drive gas turbines reject on the order of 60 % of fuel lower heating value in exhaust at 425 °C to 550 °C [Anonymous, 2024]. Treating units simultaneously require process heat. Aftercoolers reject low-grade heat. The site values on-stream factor highly. Any recovery scheme that adds backpressure, water plant, or trip risk is discounted.

LNG is therefore a severe test: high-grade residual heat exists, but so does a mandatory process-heat customer and a low tolerance for novelty.

5.2 Mapped cascade

Table 1: RHC mapping on a gas-turbine-driven LNG train.

Stream	Typical source after prior stage	LNG use
1	GT exhaust as received	Hot oil / steam for acid-gas and water removal
2	Exhaust after process coil	Partial-heat sCO ₂ or ORC electricity
3	Cooler flue gas; refrigerant aftercoolers	Absorption inlet chilling; optional LP steam
4	Surplus of Streams 2–3 at part load	Concrete / packed bed + hot-water tank
—	Cold-box JT valves (optional)	Two-phase expander, more liquid

5.3 Expected character of the gain

Integration studies that add absorption refrigeration or ORC to gas-turbine exhaust on C3MR and cascade processes have reported liquefaction power reductions on the order of 15 % to 21 % and, for a roughly

1.4 Mtpa example, several thousand tonnes of CO₂ per year avoided [He and Lin, 2020]. A steam bottomer on a published multi-turbine layout recovered on the order of 46 MW_e; an ORC on the same class of problem recovered on the order of 22 MW_e [Anonymous, 2024]. Two-phase expanders in place of JT valves have been quoted at about 5 % to 7 % additional liquid [Elliott Group, 2024].

RHC does not add those figures as if they were independent and simultaneous. Process-first operation will reduce Stream 2 heat below a design that takes the whole exhaust. Inlet chilling increases product and also changes turbine exhaust temperature and flow. The honest claim is directional: residual stack heat and aftercooler heat can be converted into a mix of megawatts, extra tonnes, and displaced fired duty, provided backpressure is capped and the process coil is not starved.

5.4 Electric-drive plants

On eLNG trains the refrigerant compressors do not reject GT exhaust. The waste-heat problem moves to the power station. If that station is already combined cycle, RHC on the liquefaction plot is mostly aftercooler heat pumps and expander retrofits. If the power island is simple cycle, RHC belongs there.

6 Application beyond LNG

The sequence—process first, residual work, upgrade or chill, store—is portable. Hardware changes with dirtiness and temperature.

6.1 Close copies

Gas-pipeline compressor stations. Aero-derivative or industrial turbines drive compressors with little on-site steam demand. Stream 1 may be thin. Stream 2 (modular sCO₂ or ORC) plus Stream 3 inlet chilling is nearly a drop-in. This is the nearest commercial analogue to LNG drivers.

Refineries and petrochemicals. Fired heaters and reformers produce hot flue gas while steam headers remain in deficit. Stream 1 is air preheat and steam. Stream 2 is residual power. Stream 3 is heat-pump steam for low-pressure users. Storage absorbs turnarounds and swing units.

CHP and simple-cycle generators. An HRSG already implements Stream 1–2 in steam form. RHC asks whether stack gas after the HRSG and jacket or condensate loops still justify a heat pump or a small ORC, and whether a buffer can keep the tail end online.

6.2 Dirty high-temperature stacks

Cement kilns, steel furnaces, glass tanks, and incinerators reject very hot, dusty, sometimes corrosive gas. Stream 2 may be a packed-bed or ceramic store feeding Rankine or ORC rather than a compact printed-circuit sCO₂ heater. Powder hold-up and acid dew point dominate design. The cascade rule still holds: clinker coolers and boiler coils come before power, and a packed bed is both store and heat exchanger.

6.3 Continuous mid-grade steam users

Pulp and paper, many chemical plants, and textiles have 24-hour steam demand. Stream 3 heat pumps and Stream 1 process reuse will often outrun Stream 2. RHC still forbids installing an ORC on a stream that can displace a boiler.

6.4 Low-grade sites

Food and beverage plants, pharmaceuticals, and data centres reject heat at 30 °C to 90 °C. Stream 2 is usually omitted. The cascade collapses to reuse, heat pump, tank, and—where pipes exist—district offtake. That is still RHC: quality matching without a vanity turbine.

6.5 Marine engines

Exhaust, jacket water, and scavenge-air coolers are already three grades. Hotel load and propulsion load do not always coincide. A small store plus absorption cooling or ORC on exhaust is the marine form of Streams 3 and 4.

Table 2: Which RHC streams dominate by sector.

Sector	S1 process	S2 power	S3 upgrade/chill	S4 hours-store
LNG mechanical drive	High	High	High (inlet chill)	Medium
Pipeline compressors	Low–medium	High	Medium	Low
Refining / petrochemicals	High	Medium	High	Medium
Cement / steel / glass	High	Medium	Low–medium	High
Pulp, paper, chemicals	High	Low–medium	High	Medium
Food / data centres	Medium	Low	High	High
Marine	Medium	Medium	Medium	Medium

7 Comparison with single-device strategies

Table 3: Qualitative comparison on a GT-driven process plant.

Strategy	Strength	Failure mode	RHC response
Process only	Simple, familiar	Residual stack unused	Adds S2–S4 on remainder
Steam combined cycle	Highest large-GT work	Water, footprint, availability	Allowed as S2 if owner accepts
Recuperated sCO ₂	Compact, dry, high η_{cyc}	Hot stack, fights process coil	Partial-heat layout after S1
ORC on aftercoolers	Commercial package	Tiny Carnot, missed steam duty	Move that stream to S3
Heat pump only	Excellent low-grade use	Leaves high-grade work unused	Heat pump is S3, not the whole train
Store only	Handles mismatch	Does not allocate quality	Store is S4, not a substitute for S1–S3

The architectural claim can be stated as follows. Let Q_1 , Q_2 , and Q_3 be heat quantities whose exergy decreases in that order. A single device forced to swallow $Q_1 + Q_2 + Q_3$ will be oversized for the low-grade part, undersized in temperature for the high-grade part, or both. Three devices without a priority rule will bid for Q_1 and leave Q_3 on the ground. RHC is the priority rule plus the buffer that keeps the lower stages from being switched off.

8 Implementation

8.1 Packaging

The intended commercial form is not a unique turbomachine. It is a specified interface:

- flange and backpressure guarantee on the host driver or furnace;
- process-coil first, with hard interlock;
- a catalogue Stream 2 block ($s\text{CO}_2$ or ORC) sized on residual—not raw—exhaust;
- a catalogue Stream 3 block (heat pump or absorption);
- a catalogue Stream 4 module (concrete, packed bed, or water);
- one supervisory controller with fail-safe bypass.

That packaging matches how industrial buyers actually purchase: availability numbers, not papers.

8.2 Materials and dirty gas

Printed-circuit and compact exchangers belong on clean turbine exhaust. Packed beds, heat-pipe economizers, and washable condensing economizers belong on dusty or wet flue gas. Deep latent-heat recovery below acid dew point is a materials problem before it is a cycle problem. RHC does not remove that problem; it confines expensive corrosion-resistant surface to the stream that actually needs it.

8.3 Business model

Heat-as-a-service fits Streams 3 and 4 particularly well: a third party owns the skid and sells steam or chilled water from savings. Stream 2 on a production-critical driver will more often be owner-operated because trip risk sits on the host shaft. Metering of dumped heat is a precondition. Unmetered stacks do not get budgets.

8.4 Design-for-retrofit

New plants should leave spare plot, stack flanges, and backpressure margin even if Stream 2 is deferred. Retrofit cost is a principal reason existing heat stays dumped.

9 Limits and failure modes

RHC will not:

- replace the host process (liquefaction still needs compressors; cement still needs a kiln);
- double thermal efficiency of a simple-cycle driver without accepting combined-cycle bulk;
- make electricity from 40 °C loops at attractive cost;
- survive a design that places the power heater upstream of the process coil;
- survive unbounded exhaust backpressure;
- remain in service if there is no buffer and operators isolate it at every turndown;
- create a district-heat customer where no pipe and no offtaker exist.

Thermoelectrics and exotic low- ΔT generators are outside the default cascade. They may serve sensors or very small parasitic loads. They are not the megawatt path.

The concept is integration. Claims of novelty should be limited to the priority-and-buffer architecture and to waste-heat-specific power layouts, not to the invention of heat exchangers or turbines.

10 Research and deployment agenda

Near-term work that would make RHC bankable:

1. Published residual-exhaust maps for common mechanical-drive turbines after a specified process-coil duty, including backpressure sensitivity.
2. Side-by-side optimization of partial-heating $s\text{CO}_2$ versus ORC on those maps, maximizing $\dot{W}_{\text{net}}$ in (2), not η_{cyc} .
3. Inlet-chilling versus power trade-off curves versus ambient temperature for LNG and pipeline drivers.
4. Dirty-gas packed-bed plus heat-pump tails for cement and steel.
5. Open supervisory-control specification so vendor islands can be sequenced without a custom project.
6. Two reference retrofits: one clean GT driver, one dirty furnace stack, each with metered dumped heat before and after.

11 Conclusions

Industry does not lack waste-heat devices. It lacks an allocation rule that keeps those devices from fighting the host process and from being bypassed when load changes. The Residual Heat Cascade is that rule made into a modular train: process heat first, residual work second, upgraded heat or cooling

third, short-term storage fourth. On gas-turbine-driven LNG plants the architecture converts leftover exhaust and aftercooler heat into a combination of electricity, inlet-chilling tonnes, and displaced fired duty, without claiming a new liquefaction process. The same sequence applies to compressor stations, refining, furnace industries, steam-using factories, ships, and low-grade sites—with Stream 2 omitted where Carnot so requires.

Better efficiency and more usable power, in this framing, are the result of not spending high-grade heat on low-grade jobs and not spending low-grade heat on engines. That is an engineering programme, not a pending discovery.

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